IV Infusions Practice 3

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

b A female patient (35 years old, 65 kg) is to be given a drug by IV infusion. The elimination half-life is 7 hours and the apparent volume of distribution is 23.1% of body weight. The desired steady-state plasma level is 10 mcg/mL. What is an appropriate infusion rate?

a A female patient (35 years old, 65 kg) with normal renal function is given a drug by IV infusion with no loading dose. The elimination half-life is 7 hours. How long after the start of the infusion would it take to reach 95% of the steady-state concentration?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

b

Her key

Answer: 14.86 mg/hr

WHAT THE STEM GIVES

age	35 years old	*not needed*
weight	65 kg	with the percentage gives $V_D = 0.231 \times 65 =$ 15.015 L
t½	7 hours	gives $k = \frac{0.693}{7} = 0.099\ hr^{-1}$
V_D	23.1% of body weight	the fraction of 65 kg that is the volume
C_{ss}	10 mcg/mL	the target, as 10 mg/L, multiplied by $k \cdot V_D$

THE WORKING, ONE LINE AT A TIME

SETUP

$R = C_{ss} \cdot Cl = C_{ss} \cdot k \cdot V_D$, so a rate constant and an absolute volume both have to be built before the multiplication

The rate that holds a chosen plateau is that plateau times clearance, because at steady state the amount delivered each hour must equal the amount removed. The stem gives a half-life and a percentage of body weight, not a clearance, so those are assembled first, in matching units.

ALGEBRA

V_D = 23.1% of 65 kg = $0.231 \times 65 = 15.015$ L

Applying the percentage to the body weight is an evaluation, not a unit conversion. The step from kilograms to litres assumes one kilogram of body weight holds one litre of distribution volume; Dr. Mosley uses this throughout without flagging it, and the answer depends on it entirely.

UNIT

$C_{ss} = 10$ mcg/mL = 10 mg/L

Micrograms per millilitre and milligrams per litre are numerically identical, as she instructs the class to take for granted rather than converting. Working in milligrams per litre keeps the final rate in milligrams per hour.

ALGEBRA

$k = \frac{0.693}{7\ hr} = 0.099\ hr^{-1}$

The clearance needed for the rate is the product of the rate constant and the volume of distribution, so the half-life is converted first.

ALGEBRA

$R = C_{ss} \cdot k \cdot V_D = (10\ mg/L)(0.099\ hr^{-1})(15.015\ L) = 14.86\ mg/hr$

The implied clearance is 1.486 L/hr, and 10 multiplied by that gives 14.86. The litres cancel between the volume and the concentration, leaving milligrams per hour.

ROUND

R = 14.86 mg/hr, recommended as 15 mg/hr

Her key prints 14.86 mg/hr and then rounds it to 15 mg/hr as the practical recommendation, writing the two with an arrow between them. Either value falls inside the tolerance here.

> **DOES THE NUMBER MAKE SENSE?**
> V_D is 23.1% of 65 kg, 15.015 L, and k is 0.693 over 7, 0.099 hr^{-1}, so Cl is about one and a half L/hr; 10 mg/L times that is about 15 mg/hr, and 14.86 fits.

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a

Her key

Answer: 30.3 hr

WHAT THE STEM GIVES

age	35 years old	*not needed*
weight	65 kg	*not needed*
t½	7 hours	gives $k = \frac{0.693}{7} = 0.099\ hr^{-1}$
fraction of C_{ss}	95%	so $e^{-kt} = 0.05$ and $kt = -\ln(0.05) = 3.00$

THE WORKING, ONE LINE AT A TIME

SETUP

$0.95 = 1 - e^{-kt}$, so $e^{-kt} = 0.05$

Reaching 95 per cent of steady state means the fraction bracket equals 0.95, leaving 5 per cent of the gap uncovered. The patient's weight and volume of distribution play no part, because the plateau cancels out of a percentage question.

ALGEBRA

$kt = -\ln(0.05) = 3.00$

Taking natural logarithms of both sides gives 2.9957, which she rounds to 3. This is the same reason 3 divided by k is often quoted as a shortcut for the time to 95 per cent of steady state.

ALGEBRA

$$k = \frac{0.693}{7\ hr} = 0.099\ hr^{-1}$$

The rate constant is what converts the dimensionless product kt into a time. Dividing 0.693 by 7 hours leaves reciprocal hours.

ALGEBRA

$$t = \frac{3}{0.099\ hr^{-1}} = 30.3\ hr$$

A pure number divided by reciprocal hours gives hours. Her second route, 4.32 half-lives multiplied by 7 hours, gives 30.24 hr, the small difference coming from her rounding of 4.32 rather than from any difference of method.

ROUND

t = 30.3 hr

Her key prints 30.3 hr, to one decimal place. The answer is a time and carries no reciprocal unit; the age and weight given in the stem never enter it, because the plateau cancels out of any question asked as a percentage of steady state.

DOES THE NUMBER MAKE SENSE?

95 per cent of the plateau is reached between four and five half-lives of 7 hours, so t lies between 4 × 7 and 5 × 7 hr; 30.3 hr fits.
